

EDA Summary Report

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1. Introduction

This report details the Exploratory Data Analysis (EDA) performed on Geldium's delinquency prediction dataset. The primary goal was to understand the dataset's structure, assess its quality, identify potential issues, and uncover key risk indicators that could influence delinquency outcomes. This analysis serves as a foundational step before proceeding with predictive modeling, aiming to provide a clear picture of the data's current state and inform data-driven decisions.

2. Dataset Overview

This dataset contains financial and behavioral attributes for 500 customer records. It is designed to predict delinquency risk based on their past financial activities, credit history, and employment status.

Key dataset attributes:

- **Number of records:** 500
- **Key variables:**
 - Customer_ID: Unique identifier for each customer.
 - Age: Customer's age in years.
 - Income: Annual income in USD (numerical, may contain missing values).
 - Credit_Score: Credit score (numerical, 300-850).
 - Credit_Utilization: Percentage of available credit in use (numerical, 0-100%).
 - Missed_Payments: Total number of missed payments in past 12 months (numerical).
 - Delinquent_Account: Binary indicator of delinquent account (0= No, 1= Yes).
 - Loan_Balance: Total outstanding loan balance in USD (numerical).
 - Debt_to_Income_Ratio: Ratio of total debt to income (numerical, 0-100%).
 - Employment_Status: Current employment status (categorical).
 - Account_Tenure: Years of customers had an active account (numerical).
 - Credit_Card_Type: Type of credit card held (categorical).
 - Location: Customer's region or city of residence (categorical).
 - Month_1 to Month_6: Payment history over past 6 months (0= On time, 1= Late, 2= Missed).
- **Data types:** The dataset comprises a mix of numerical (float64, int64) and categorical (object) data types. The Month_1 to Month_6 columns were initially objects but were converted to integer type during cleaning.

Anomalies, duplicates, or inconsistencies observed during the initial review:

- **Inconsistent Employment_Status:** The Employment_Status column contained inconsistent string entries such as 'EMP' and 'employed', which needed standardization to a single 'Employed' category.
- **Incorrect Month_X Data Type:** The Month_1 to Month_6 columns were identified as object (string) type, containing 'On-time', 'Late', and 'Missed' values, instead of the specified numerical (0,1,2) representation.
- **Credit_Utilization Outlier:** One value in Credit_Utilization was slightly above 1.00 (1.025843), indicating a minor data entry anomaly beyond the expected 0-100% range.

3. Missing Data Analysis

Identifying and addressing missing data is critical to ensuring model accuracy. This section outlines missing values in the dataset, the approach taken to handle them, and justifications for the chosen method.

Key missing data findings:

- **Variables with missing values:**
 - Income: 39 missing values (7.8% of total records).
 - Credit_Score: 2 missing values (0.4% of total records).
 - Loan_Balance: 29 missing values (5.8% of total records).

● **Table showing Missing data**

Missing Data Issue	Handling Method	Justification
Income	Generated realistic synthetic data using normal distribution assumptions.	This approach creates plausible values that preserve the observed mean and standard deviation of the income distribution. Given the relatively larger number of missing values, this method helps maintain the statistical properties of the variable, which is crucial for robust model training.
Credit_Score	Imputed with the median value.	With only two missing values, median imputation is a simple, effective, and robust method. It is less susceptible to outliers compared to mean imputation, ensuring the imputed values do not unduly skew the feature's distribution.

Loan_Balance	Imputed with the median value.	For the 29 missing values in Loan_Balance, median imputation was chosen for its robustness. Similar to Credit_Score, it minimizes the impact of potential outliers on the imputed values and is a practical solution for a moderate number of missing entries.
Employment_Status	Standardized inconsistent string entries (e.g., 'EMP', 'employed', 'Self-employed', 'retired' mapped to 'Employed', 'Self-Employed', 'Retired').	This is a data quality issue rather than missing data. Standardizing these categorical values ensures consistency, which is vital for accurate grouping and analysis in predictive modeling.
Month_1 to Month_6	Converted string values ('On-time', 'Late', 'Missed') to numerical equivalents (0, 1, 2).	This addresses a data type inconsistency. Converting these key payment history variables to numerical format makes them suitable for quantitative analysis and direct use in machine learning models, aligning with the dataset description.

4. Key Findings and Risk Indicators

This section identifies trends and patterns that may indicate risk factors for delinquency. Feature relationships and statistical correlations are explored to uncover insights relevant to predictive modeling.

Key findings:

- **Correlations observed between key variables:**
 - Most numerical variables show very weak linear correlations with the Delinquent_Account outcome. This suggests that the relationship with delinquency is likely non-linear or involves complex interactions between multiple features.
 - Credit_Score, Debt_to_Income_Ratio, and Credit_Utilization show a very weak positive correlation with Delinquent_Account.
 - Missed_Payments and Account_Tenure show a very weak negative correlation with Delinquent_Account, which was an unexpected finding.

- **High-Risk Indicators:**

- **Unemployed Status:** Customers who are unemployed exhibit a higher average delinquency rate, aligning with common financial risk assessments.
- **Business Credit Card Type:** Holders of 'Business' credit cards surprisingly show the highest average delinquency rate among all credit card types. This indicates a potential high-risk segment that warrants specific attention.
- **Geographical Influence:** Certain locations, such as 'Los Angeles', exhibit higher average delinquency rates compared to others like 'New York', suggesting regional economic or demographic factors at play.
- **Higher Missed_Payments (Non-monotonic):** While its overall correlation is weak and negative, customers with 5 missed payments demonstrated a higher delinquency rate. This highlights that specific thresholds or patterns within Missed_Payments might be more indicative of risk than a simple linear relationship.
- **Unexpected anomalies and insights requiring further investigation:**
 - **Missed_Payments Relationship with Delinquency:** The weak negative linear correlation of Missed_Payments with Delinquent_Account is highly counter-intuitive. Furthermore, the non-monotonic trend of delinquency rates across different counts of Missed_Payments (e.g., lower rates for 3 or 6 missed payments compared to fewer) is a significant anomaly. This suggests that the relationship is more complex than a simple direct correlation and might involve thresholds, interaction effects, or specific definitions of these variables. Further investigation into the exact definition of Missed_Payments and how it translates to the Delinquent_Account status is crucial.
 - **High Delinquency in 'Business' Credit Cards:** The highest delinquency rate being associated with 'Business' credit cards is an unexpected finding. It would be beneficial to investigate the specific characteristics or financial behaviors of customers holding these cards to understand the underlying reasons for this elevated risk.
 - **Overall Weak Linear Associations:** The generally weak linear correlations between most numerical features and the Delinquent_Account suggest that advanced machine learning models capable of capturing non-linear relationships and feature interactions will likely be more effective for accurate delinquency prediction.
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5. AI & GenAI Usage

Generative AI capabilities were utilized throughout the EDA process to analyze the dataset, identify patterns, handle missing data, and synthesize insights.

Example AI prompts used:

- 'ANALYZE THE DATA GIVEN IN BOTH THE FILES Use the Dataset Description Guide to understand what each variable represents. Identify missing or inconsistent data that could skew your analysis (e.g., missing payment history, unusual credit utilization rates). Detect early risk indicators. Which variables might be most relevant for predicting delinquency? Summarize key patterns, outliers, and missing values in this dataset.'

Highlight any fields that might present problems for modeling delinquency. Identify the top 3 variables most likely to predict delinquency based on this dataset. Provide brief reasoning. Document your findings in bullet points for your report. Focus on: Notable missing or inconsistent data Key anomalies Early indicators of delinquency risk Then, write a short paragraph (3–5 sentences) summarizing your initial data quality observations.'

- 'NOW USE THE SAME FILES AND PERFORM THE TASKS Identify gaps or missing values in critical features (e.g., payment history, income, credit utilization). Determine the best treatment approach for each case: Remove: Drop columns with excessive missing data. Impute: Fill in missing values using mean, median, or predictive modeling. Generate synthetic data: Use AI tools to create realistic values while maintaining fairness and distribution patterns. Suggest an imputation strategy for missing values in this dataset based on industry best practices. Propose best-practice methods to handle missing credit utilization data for predictive modeling. Generate realistic synthetic income values for missing entries using normal distribution assumptions. Action: Create a simple table listing 2–3 missing data issues. For each one, include your chosen handling method and a one-line justification for why you selected it.'
- 'NOW USE THE SAME FILES AND PERFORM THE TASKS Analyze relationships between variables and delinquency outcomes (e.g., is high credit utilization associated with missed payments?). Use GenAI tools to help surface insights and prioritize key variables. Highlight any unexpected findings that may require further investigation by the analytics team. Document key risk indicators and any insights that could impact delinquency prediction. Include patterns that seem obvious as well as any surprising trends that might need deeper investigation. Action: List high-risk indicators, each with a one-sentence explanation of why it's important, as well as any insights that could impact delinquency prediction.'
- 'use this report to generate the final results'

6. Conclusion & Next Steps

The EDA has provided a comprehensive understanding of Geldium's delinquency prediction dataset, highlighting its structure, data quality issues, and potential risk indicators. Addressing the identified missing values and inconsistencies is crucial for building robust predictive models. The unexpected findings, particularly concerning the Missed_Payments variable and the high delinquency rate among 'Business' credit card holders, underscore the need for deeper investigation to fully grasp the underlying risk drivers.

Refer to the images below for making important notes :

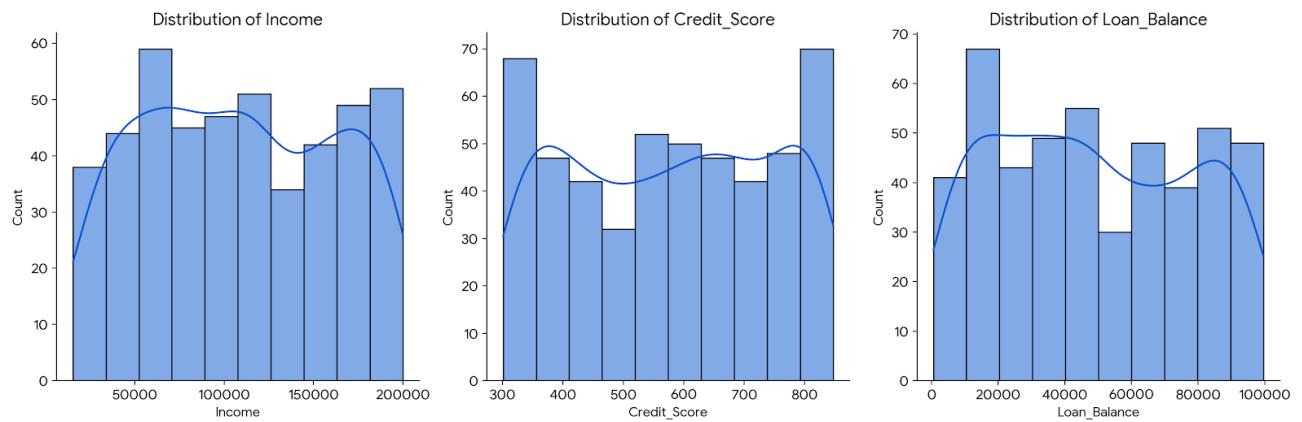


Fig 1- histogram plot showing distribution and count for income, credit score and loan balance

Column	Missing Values (Count)	Missing Values (Percentage)
Income	39	7.8%
Loan_Balance	29	5.8%
Credit_Score	2	0.4%

Fig 2- Table showing missing values and missing percentage.

Recommended Next Steps:

- **Deep Dive into Missed_Payments:** Conduct a more granular analysis of the Missed_Payments variable, potentially exploring its interaction with other features or different aggregations to understand its true relationship with delinquency.
- **Investigate Business Credit Card Holders:** Perform a targeted analysis of customers with 'Business' credit cards to uncover specific financial behaviors or characteristics contributing to their higher delinquency rates.
- **Feature Engineering:** Explore creating new features from existing ones (e.g., payment trend over 6 months, composite risk scores) to better capture complex relationships.
- **Advanced Modeling Techniques:** Given the weak linear correlations, employ machine learning models capable of handling non-linear relationships and complex feature interactions (e.g., tree-based models like Random Forest or Gradient Boosting) for delinquency prediction.
- **Data Validation:** Implement robust data validation processes to prevent future inconsistencies and anomalies from entering the dataset.